Practical work – Stress detection from physiological signals (WESAD)

Course: Applied Machine Learning for Real-World Data (AML)
Academic year: 2026–2027
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Context

In this practical work you will build a machine-learning pipeline that detects **stress** from wearable physiological signals. The data come from the public **WESAD** dataset (Schmidt et al., 2018), in which 15 subjects wore a chest-worn device (ECG, EDA, respiration, ...) while going through different conditions:

- **Base** – neutral reading task (baseline);
- **TSST** – Trier Social Stress Test (public speaking and mental arithmetic): the **stress** condition;
- **Fun** – amusement condition (funny video clips).

The raw signals have already been processed for you: they were cut into **60 s windows** (one window every 10 s) and a set of **32 features** was extracted from each window (ECG/HRV, EDA, respiration). Your work starts from this feature table.

Goals

- explore a real-world physiological dataset and its quality issues (missing values, class imbalance, inter-subject variability);
- understand the provided preprocessing and validation strategy, and the risks of **data leakage**;
- tune a provided random forest, implement at least one other classifier, and compare them with a baseline for stress detection;
- evaluate honestly (subject-independent validation, appropriate metrics) and **interpret** the results from a physiological point of view.

AI use and academic integrity policy

- **You might be asked to explain your work.** You may be asked to explain any part of your code, justify a choice (imputation, normalisation, model, hyperparameters, metric, validation scheme) and interpret your results.
- **Answers must be in your own words** and supported by your own plots and numbers.

The dataset

The file `stress_detection_dataset.csv` (provided in this folder) contains one row per window.

Column(s)	Description
`subject`	Subject identifier (15 subjects)

Column(s)	Description
`window_id`, `window_start`, `window_end`, `window_duration_sec`	Position of the window in the recording (not features!)
`condition`	`Base`, `TSST` or `Fun`
`label`	Numeric label: 0 = Base, 1 = TSST, 2 = Fun
`ECG_Rate_Mean`, `HRV_*`	Heart rate and heart-rate-variability features (time and frequency domain) -> Stress generally increases heart rate and reduces heart-rate variability due to changes in autonomic nervous system activity.
`EDA_*`, `SCL_*`, `SCR_*`	Electrodermal activity -> Stress activates sweat glands, increasing skin conductance.
`RSP_*`, `RRV_MeanBB`	Respiration features and respiratory rate variability -> Stress can cause faster, shallower breathing and changes in respiratory variability.

Things to keep in mind:

- some features may contain **missing values** (e.g. when a window contains too few detected peaks);
- the classes are **not balanced**, and subjects do not contribute equally;
- consecutive windows are 10 s apart but still overlap by 50 s: they remain **very similar**.

1) Exploration

Understand your data before modelling. At a minimum:

- explore class distribution and missing values. Are the classes balanced, overall and for each subject? Where are the missing values concentrated? What do these observations imply for the preprocessing?
- compare the distribution of several features between **Base** and **TSST** (eg boxplots);
- check the correlation between features. Which groups of features are redundant?

Formulate an initial hypothesis: which features or modalities (ECG, EDA, or RESP) do you expect to distinguish stress from baseline most effectively in this experiment, and why?

2) Preprocessing

The preprocessing is **provided** in the notebook. Read it carefully and be able to discuss its choices:

- the columns that are not features are discarded;
- windows with missing values are dropped;
- the features are normalised **per subject** (z-score with the mean and standard deviation of each subject's own windows);
- the class imbalance is handled with class weights and by reporting balanced accuracy.

Discuss in particular:

- the pros and cons of a per-subject normalisation, in particular in a real-world deployment (what do you need from a new subject?);
- the advantages, drawbacks and alternatives of dropping the windows with missing values;
- why class weights and balanced accuracy are preferable to accuracy alone.

> **Warning – data leakage.** Any statistic computed from the data (imputation values, mean/std for scaling, selected features) must be fitted on the **training set only**. Keep this in mind for any step you add yourself.

3) Modelling

For the binary problem **Base vs TSST**, a **random forest** is provided as a starting point. You must:

1. **tune** hyperparameters of the random forest;
2. **implement at least one other classifier** (e.g. logistic regression, SVM, k-NN, gradient boosting, MLP) and tune one or two of its hyperparameters;
3. add a simple **baseline** for context;
4. **compare** all the models.

Validation strategy

Because windows overlap and come from the same subject, a random split puts nearly identical samples in both the training and validation sets and gives **over-optimistic** results. A **leave-one-subject-out (LOSO)** scheme is provided in the notebook. You must:

- explain what leave-one-subject-out is and what are its advantages;
- explain why selecting hyperparameters with the same LOSO scores that you report is slightly optimistic, and how you could prevent it.

4) Evaluation

- Report several metrics (e.g. accuracy, balanced accuracy, F1, ROC-AUC) and justify which one matters most here;
- build one summary table comparing all your models;
- show a confusion matrix for your best model;
- report the **per-subject** performance for your best model: is the model equally good for all subjects?

5) Interpretation

- Which features are the most useful ? Do they agree with your initial hypothesis (section 1)?
- Which **modality** seems to be the most informative (ECG: cardiac activity, EDA: perspiration, respiration)? How would you interpret it from a physiological point of view ?
- Analyse the errors: which subjects are misclassified, why could that be ?
- Discuss briefly the limitations of your system and whether you would trust it in the real world.

6) Extension (optional) – adding the *Fun* condition

For students who finish early. The dataset also contains the **Fun** condition.

1. Turn the problem into a **three-class** classification (Base / TSST / Fun). Re-evaluate your best model(s) and discuss the confusion matrix.
2. Which classes are confused and why? Which features help to separate stress from amusement?

Bonus) Feature selection (optional)

Reduce the feature set with a method of your choice (filter, wrapper, embedded, ...) and study the effect on performance.

> **Note** Feature selection must use the **training set only**, inside each cross-validation fold.

Deliverables

1. The completed notebook `AML_StressDetection_PW2627.ipynb` (all `TODO` cells and written-answer cells of sections 1 to 5 completed, all questions answered in your own words; section 6 and the bonus are optional).
2. A short **oral defence** during which you will be asked to explain your code, your choices and your results.

Reference

Schmidt, P., Reiss, A., Duerichen, R., Marberger, C., & Van Laerhoven, K. (2018). *Introducing WESAD, a multimodal dataset for wearable stress and affect detection.* ICMI 2018.